Honors Calculus III, Homework 3

Please upload solutions to **Questions 1, 2(b) and 5** by **8pm on Friday 14**. You can use any results you want from lectures, provided you state them clearly, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says "optional". As usual, for full credit you need to show and explain your work.

1. (4pt) Let $v, w \in \mathbf{R}^n$ be nonzero vectors. Compute the norm of the vector $||v||w - ||w||v$ and deduce a new proof of the Cauchy–Schwarz inequality.

2. (a) Let $x, y \in \mathbf{R}^n$. Prove the *polarization identity*:

$$\langle x, y\rangle = \frac{1}{4}(||x+y||^2 - ||x-y||^2).$$

   (b, 4pt) We say that a function $f : \mathbf{R}^n \to \mathbf{R}^n$ preserves lengths if $||f(x)|| = ||x||$ for all $x \in \mathbf{R}^n$. We say that f preserves angles if for all nonzero $x, y \in \mathbf{R}^n$, the angle between $x, y \in \mathbf{R}^n$ coincides with the angle between $f(x)$ and $f(y)$. Prove that if f is linear and preserves lengths then it preserves angles.

   (c) Give an example of a linear map that preserves angles but does not preserve lengths.

3. Prove that the coordinate functions $c_i : \mathbf{R}^n \to \mathbf{R}$ are linear for all $1 \le i \le n$. If $f : \mathbf{R}^m \to \mathbf{R}^n$ is a function, prove that f is linear if and only if $f_i = c_i \circ f$ is linear for all $1 \le i \le n$.

4. For each of the following functions, say whether or not it is linear (no proof needed). If it is, compute its matrix in the standard basis.

   (a) $f : \mathbf{R} \to \mathbf{R}, f(x) = \sin(x)$.
   (b) $f : \mathbf{R}^4 \to \mathbf{R}, f(x, y, z, w) = x + 2y + 3z + 4w$.
   (c) $f : \mathbf{R}^3 \to \mathbf{R}^3, f(x, y, z) = (1 + x, 1 + y, 1 + z)$.
   (d) $f : \mathbf{R}^2 \to \mathbf{R}^2, f(x, y) = (2x + 3y, 3x + 4y)$.
   (e) $f : \mathbf{R}^2 \to \mathbf{R}^3, f(x, y) = (x - y, x + y, 2x + 3y)$.

5. (4pt each) Consider the following linear maps (you do not need to prove that they are linear):

$$\begin{aligned} f : \mathbf{R}^3 \to \mathbf{R}^2, f(x, y, z) &= (x - y, y - z) \\ g : \mathbf{R}^2 \to \mathbf{R}^3, g(x, y) &= (x + y, x - y, x) \\ h : \mathbf{R}^3 \to \mathbf{R}, h(x, y, z) &= x + y + z. \end{aligned}$$

   Compute the matrix of $g \circ f, h \circ g$ and $f \circ g$.

6. Practice matrix multiplication. Here are some examples to get your started:

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\cdots\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ (n factors).}$$

You can find similar problems in Treil, *Linear algebra done wrong*, freely available at `https://www.math.brown.edu/streil/papers/LADW/LADW_2017-09-04.pdf`. For example, try Chapter 1, Exercises 3.1, 3.3(a, b), 5.1. The notation A^T means *transpose* of a matrix, that is to say forming a new matrix by swapping the rows of A with the columns of A.

7. (Optional) Prove that the following properties hold for $\mathbf{C}$ with the addition $+$ and multiplication $\cdot$ defined in class, and with $0 = (0,0), 1 = (1,0)$.

   (a) $x + (y + z) = (x + y) + z$ for all $x, y, z \in \mathbf{C}$.

   (b) $x + 0 = 0 + x = x$ for all $x \in \mathbf{C}$.

   (c) For all $x \in \mathbf{C}$ there exists $-x \in \mathbf{C}$ such that $x + (-x) = (-x) + x = 0$.

   (d) $x + y = y + x$ for all $x, y \in \mathbf{C}$.

   (e) $x(yz) = (xy)z$ for all $x, y, z \in \mathbf{C}$.

   (f) $xy = yx$ for all $x, y, z \in \mathbf{C}$.

   (g) $1 \cdot x = x \cdot 1 = x$ for all $x \in \mathbf{C}$.

   (h) $x(y + z) = xy + xz$ for all $x, y, z \in \mathbf{C}$.

   (i) For all $x \in \mathbf{C}, x \neq 0$ there exists $x^{-1} \in \mathbf{C}$ such that $xx^{-1} = x^{-1}x = 1$.

   By definition, the first four properties mean that $(\mathbf{C}, +, 0)$ is an *abelian group*. The first eight properties mean that $(\mathbf{C}, +, \cdot, 0, 1)$ is a *commutative ring with 1*. Taken together, these nine properties mean that $(\mathbf{C}, +, \cdot, 0, 1)$ is a *field*.

Solutions to graded questions. (1) Let $z = ||v||w - ||w||v$. We compute

$$(z, z) = ||v||^2||w||^2 - 2||v||||w||(v, w) + ||w||^2||v||^2.$$

Since $(z, z) = ||z||^2$ is always nonnegative, we find that

$$2||v||||w||(v, w) \leq 2||v||^2||w||^2.$$

Dividing through by the positive number $2||v||||w||$, we obtain that

$$(v, w) \leq ||v||||w||.$$

Repeating the same argument with z replaced by $||v||w + ||w||v$ we obtain

$$||v||^2||w||^2 + 2||v||||w||(v, w) + ||w||^2||v||^2 \geq 0$$

and so

$$(v, w) \geq -||v||||w||.$$

Putting these together, we deduce that $|(v, w)| \leq ||v||||w||$.

(2a) This follows by expanding out the expression at the right-hand side:

$$(x + y, x + y) - (x - y, x - y) = 4(x, y).$$

2(b) The polarization identity implies that if f is linear and preserves lengths then f preserves the inner product. Since the angle between two nonzero vectors x and y is defined to be

$$\arccos(x \cdot y/||x||||y||)$$

we see that if f preserves lengths then f preserves angles.